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A novel human pluripotent stem cell-based gene activation system identifies IGFBP2 as a mediator in the production of hematopoietic progenitors *in vitro*

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Abstract

A major challenge in the stem cell biology field is the ability to produce fully functional cells from induced pluripotent stem cells (iPSCs) that are a valuable resource for cell therapy, drug screening and disease modelling. Here we developed a novel inducible CRISPR-mediated activation strategy (iCRISPRa) to drive the expression of multiple endogenous transcription factors important for *in vitro* cell fate and differentiation of iPSCs to haematopoietic progenitor cells and used this to identify a key paracrine role for IGFBP2 in the development of hematopoietic progenitors. We first identified nine candidate transcription factors that we predicted to be involved in blood cell emergence during development, then generated novel gRNAs directed to the transcriptional start site of these transcription factors that could also be detected during scRNAseq. iCRISPRa activation of these endogenous transcription factors resulted in a significant expansion of arterial-fated endothelial cells expressing high levels of IGFBP2 and showed that IGFBP2 remodels the metabolic activity during *in vitro* endothelial to hematopoietic transition. As well as providing fundamental new insights into the mechanisms of haematopoietic cell fate and differentiation, the broader applicability of iCRISPRa provides a valuable tool for studying dynamic processes controlling developmental events as well as for recapitulating abnormal phenotypes characterised by ectopic activation of specific endogenous gene expression in a wide range of systems.

eLife assessment

This study presents **useful** findings to inform and improve the *in vitro* differentiation of hematopoietic progenitor cells from human induced pluripotent stem cells. However, while relying on a well-characterized technical approach, the data analysis is overall **incomplete** and only partially supports the main conclusions.

Introduction

The haematopoietic system develops early during gestation through so called “waves” of hematopoietic progenitors that arise in different anatomical regions and result in the production of various progenitor and stem cells ¹⁻⁵. The precise signalling pathways leading to the production of hematopoietic stem and progenitor cells (HSPCs) during embryonic development are yet to be completely defined, posing a limitation on how to recapitulate the process *in vitro* from pluripotent stem cells.

During development, HSPCs are generated by a subset of endothelial cells, known as hemogenic endothelium ⁶⁻⁹, via the endothelial to hematopoietic transition (EHT) ¹⁰. During the EHT, endothelial cells undergo profound transcriptional remodelling whereby the expression of endothelial genes is gradually downregulated, and the transcription of the hematopoietic program is initiated ¹¹ while the cells round up and eventually detach to enter the circulation ^{12,13}.

To explore the molecular control on the development of the hematopoietic system and to address the differences with the *in vitro* differentiation of induced pluripotent stem cells (iPSCs), we compared our single cell transcriptomics analysis of *in vitro* derived hemogenic endothelium and early progenitors ¹⁴ to that of the *in vivo* HSC-primed human hemogenic endothelium ¹⁵. We then developed a novel DOX-inducible CRISPR gene activation system to assess the role of the genes that were expressed at a lower level within *in vitro*-derived cells compared to their *in vivo* counterparts. We employed single-cell RNA sequencing to track the presence of guide RNAs and we monitored the phenotypic effects of gene activation. With this experimental pipeline, we identified and functionally validated a novel role for IGFBP2, IGF Binding protein 2, in the development of *in vitro* progenitors and showed that IGFBP2 remodels the metabolic activity during *in vitro* EHT.

Results

Comparison of iPSC-derived endothelial cells with AGM AGM endothelial cells dataset identifies 9 differentially expressed transcription factors

We, and others, have shown that *in vitro* differentiation of human iPSCs is a powerful tool to model intraembryonic hematopoiesis ^{14,16-18}. To understand the molecular basis underlying the challenges associated with the *in vitro* production of blood stem and progenitor cells *in vitro* from differentiating iPSCs, we compared our scRNAseq dataset from differentiating iPSCs ¹⁴ to that of cells derived *in vivo* from the human aorta-gonad-mesonephros (AGM) region ¹⁵. We integrated the transcriptomic data of *in vitro* derived endothelial (IVD_Endo) and hematopoietic cells (IVD_HPC) with that of arterial endothelial cells (aEC), arterial hemogenic cells (aHEC) and venous endothelial cells (vEC) derived from human embryos collected between

Carnegie stage 12 and 14 (**Figure 1A** [↗](#)). To identify possible target genes that could be manipulated *in vitro* to improve iPSCs differentiation, we first determined the transcription factors marking the aHEC *in vivo* (Table 1). Then, we filtered the transcription factor genes that were expressed at lower level in the IVD_Endo and IVD_HPC. This strategy identified 9 target transcription factors *RUNX1T1*, *NR4A1*, *GATA2*, *SMAD7*, *ZNF124*, *SOX6*, *ZNF33A*, *NFAT5* and *TFDP2* (**Figure 1B** [↗](#)), whose expression was detected across other AGM datasets in the hemogenic endothelium¹⁷ [↗](#) (Supplementary Figure 1).

Development of a DOX-inducible dCAS9-SAM activation system in human iPSCs

We previously developed an all-in-one Synergistic Activator Mediator system, UniSAM, that mediates the transcriptional activation of endogenous gene expression¹⁹ [↗](#). To activate the nine target genes identified in this study, we developed a novel doxycycline-inducible SAM (iSAM) cassette targeted into the *AAVS1* locus of human iPSCs (**Figure 2A** [↗](#)). We demonstrated the activation of *RUNX1C* expression in HeLa cells and that this was correlated with the doxycycline (DOX) concentration in a linear manner (Figure S2A-B). To verify the activation in human PSCs at single-cell resolution, we employed a *RUNX1C*-GFP human embryonic stem cell (hESC) reporter cell line (Figure S2C-G). As predicted, the level of expression of the mCherry tag, the fluorescent reporter within the iSAM cassette, was proportional to the concentration of DOX (Figure S2D-E) and to the number of *RUNX1C*-GFP+ cells (Figure S2 D-F). Furthermore, the *RUNX1C* expression level, measured by the mean fluorescence intensity (MFI) of the *RUNX1C*-GFP reporter, also correlated with the concentration of DOX (Figure S2G). We then tested the iSAM cassette in the iPSC line (SFCi55) (**Figure 2B-D** [↗](#)). Only when iPSCs were transfected with iSAM and the gRNA for *RUNX1C*, in the presence of DOX, was the expression of the *RUNX1C* gene and RUNX1 protein detected (**Figure 2B-C** [↗](#)).

We then targeted the iSAM cassette into the *AAVS1* locus using a Zinc Finger Nuclease (ZFN) strategy¹⁴ [↗](#),²⁰ [↗](#),²¹ [↗](#). iPSC clones that had specifically integrated the iSAM cassette into the *AAVS1* locus were validated by genomic PCR and sequencing (Figure S2A-B). The *AAVS1* locus has been reported to be a “safe harbour” site that is resistant to epigenetic silencing and indeed, we had previously demonstrated that transgenes inserted into the *AAVS1* locus under the control of the constitutively active CAG promoter were efficiently expressed both in undifferentiated and in differentiated iPSCs²⁰ [↗](#),²² [↗](#)–²⁴ [↗](#). However, after the iSAM line had been established and cultured under self-renewal conditions, we noted a dramatic reduction in the number of mCherry+ cells in undifferentiated iSAM iPSCs upon DOX induction, that we predicted to be due to transgene-silencing of the rTTA DOX-inducible cassette (Figure S3C). To overcome this problem, we treated the iSAM iPSC line with an inhibitor of histone deacetylases (HDACs), sodium butyrate (SB), reported to have no adverse effect on iPSC maintenance²⁵ [↗](#),²⁶ [↗](#) and we also confirmed that the treatment with SB had no effect on viability and cell proliferation in our culture conditions (Figure S3D-E). A short 48-hour treatment significantly increased the number of mCherry+ cells upon DOX induction, proportional to the SB concentration (Figure S3F). We, therefore, maintained the iSAM iPSCs in the presence of SB and this fully restored the inducibility of the transgene with virtually all cells expressing mCherry in the presence of DOX (Figure S3G).

To test the effect of activating the 9 target genes on the transcriptomes of differentiating iPSC cells, we engineered the gRNAs to allow their detection within the single cell RNA sequencing pipeline²⁷ [↗](#). We inserted a capture sequence prior to the termination signal to avoid any alteration in the secondary structure of the loops thus preserving the binding of the synergistic activators of the SAM system to the stem loops of the gRNAs. Of the two capture sequences available²⁷ [↗](#), we decided to use the one that was predicted to result in fewer secondary structure alterations and this new gRNA was named 2.1 (**Figure 2E** [↗](#)). We compared the activation level achieved with the new 2.1 gRNA to that of the original 2.0 backbone using various gRNAs targeting *RUNX1C* (**Figure**

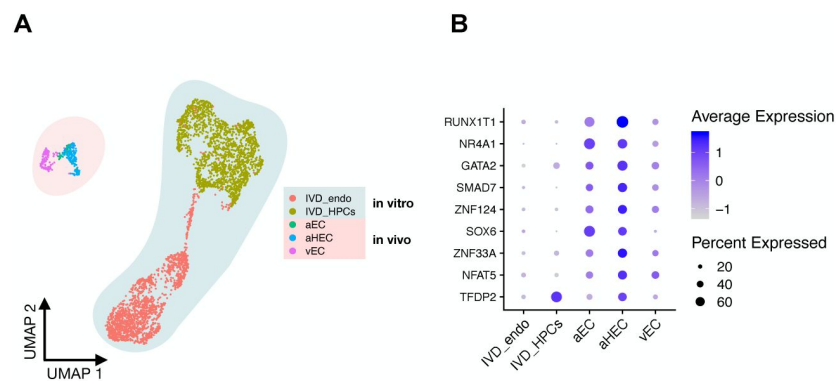


Figure 1

Comparison of *in vitro* iPSC-derived and *in vivo* AGM-derived endothelial cells identifies 9 differentially expressed transcription factors.

A - Integrative analysis of single cell transcriptome of *in vitro* derived endothelial (IVD_Endo) and hematopoietic cells (IVD_HPCs) with *in vivo* developed endothelial cells (venous, vEC; arterial, aEC; arterial hemogenic, HECs) from human embryos (CS12-CS14) visualised on UMAP dimensions. **B** - Target genes expression level showing higher expression in arterial hemogenic endothelium *in vivo* compared to *in vitro* derived cells.

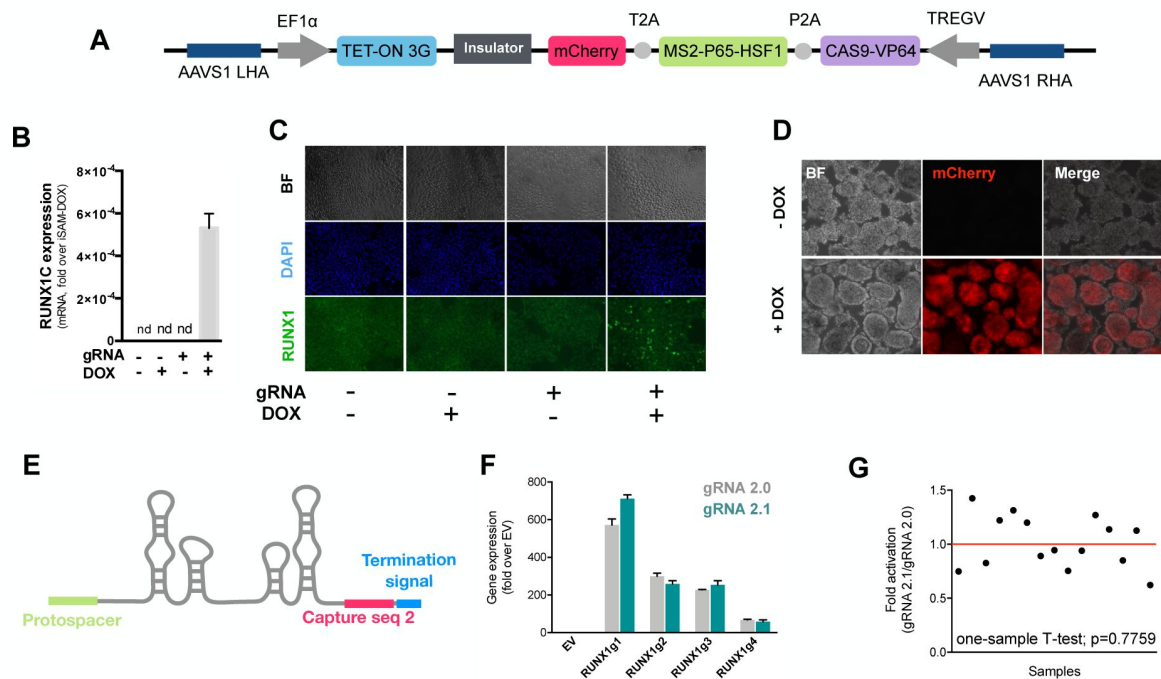


Figure 2

The inducible iSAM cassette successfully mediates activation of endogenous gene expression upon DOX induction.

A - Schematic of the iSAM cassette containing the TET-on system under the control of EF1α and dCAS9-P2A-MS2-p65-HSF1-T2A-mCherry under the rTTA responsive elements, separated by genetic silencer and flanked by AAVS1 specific homology arms. **B** - *RUNX1C* gene expression activation after transient transfection of the iSAM plasmid and gRNAs in presence or absence of DOX in human iPSC line (n=3 from independent transfections). **C** - RUNX1 protein expression upon iSAM activation after transient transfection of the iSAM plasmid and gRNAs with DOX in human iPSC line detected by immunostaining. **D** Expression of the iSAM cassette reported by mCherry tag during the differentiation protocol, the representative images (bright field - BF, and fluorescence) show embryoid bodies at day 3 of differentiation. **E** - Schematic of the gRNA 2.1 containing the capture sequence for detection during the scRNAseq pipeline. **F** - *RUNX1C* gene activation level obtained using either the gRNA 2.0 or 2.1 backbone (n=3 from independent transfections of the 4 different gRNAs). **G** - Statistical analysis of the gRNAs activation level showing no significant variation following addition of the capture sequence (n=3 for each of the 4 different gRNAs).

2F [↗](#)). These results convincingly demonstrated that the addition of the capture sequence in the gRNA 2.1 does not alter the level of endogenous gene activation that could be achieved (**Figure 2G** [↗](#)). Altogether, these results show that the iSAM system is able to induce gene expression that translates into an increased protein expression and thus provides a platform to steer phenotypical changes in cell identity.

To activate our target genes, we designed 5-7 gRNAs in the 200bp upstream of the transcriptional start sites of each of the 9 target genes. We subcloned a total of 49 gRNAs (Table S1) into the gRNA 2.1 backbone and packaged them into lentiviral particles, (herein referred to as the AGM library) as well as a non-targeting (NT) gRNA that was used as control. The iSAM iPSC line was transduced with the targeting gRNAs or the control non-targeting gRNA to generate the iSAM_AGM and iSAM_NT iPSCs line, respectively. After puromycin selection, their integration in the genome was also confirmed by PCR and sequencing (Figure S3H-I).

Single Cell RNA sequencing in combination with CRISPR activation identifies arterial cell type expansion in association with higher hematopoietic progenitor potential

To assess the transcriptional changes in response to the activation of the target genes, we differentiated the iSAM iPSCs, induced with DOX and subjected them to single-cell RNA sequencing using the 10X pipeline. After 10 days of differentiation, in the presence or absence of DOX, we FAC-sorted live CD34⁺ cells from iSAM_AGM and the iSAM_NT iPSCs. Following data filtering, we selected cells in which the gRNAs expression was detected and showed that our approach activated all the target genes, except for ZFN124. A higher level of expression of *RUNX1T1*, *NR4A1*, *GATA2*, *SMAD7*, *SOX6*, *ZNF33A*, *NEAT5*, *TFDP2* was observed following DOX treatment of cells of iSAM_AGM cells compared to the iSAM_NT cells (**Figure 3A** [↗](#)). To study the effect of gene activation on transcriptional and cellular phenotype, we performed clustering analysis and detected a total of 7 clusters (**Figure 3B** [↗](#)). A high level of *GJA4* and *DLL4* expression was used to annotate the arterial-like cell cluster, while high levels of hemogenic markers, such as *RUNX1* and *CD44*, were used to annotate the hemogenic clusters, EHT_1 and EHT_2 (**Figure 3B, C-D** [↗](#)). To assess the effect of the activation on cell identity, we analysed the proportion of the cell clusters between the libraries and detected a significant expansion of the arterial cluster in the DOX-induced iSAM-AGM compared to the iSAM_NT cells (**Figure 3E** [↗](#)). To validate the effect of the activation on the expansion of arterial cell population, we analysed their prevalence by flow cytometry. Although DOX treatment resulted in an average 1.63 ± 0.13 fold increase of CD34⁺DLL4⁺ cells in the control iSAM_NT sample (Figure S3A), the increase observed in the iSAM-AGM cells was significantly greater, with an expansion of 3.33 ± 0.73 fold increase of CD34⁺DLL4⁺ immunophenotypic arterial cells identified by flow cytometry (**Figure 3F** [↗](#)). To assess the effect of this predicted arterial cell expansion on the emergence of functional progenitors, we isolated CD34⁺ cells using magnetic beads and cocultured 20,000 cells on OP9 supportive stromal cells for 7 days, in the presence of hematopoietic differentiation cytokines. After one week, the progenitor cells were assessed by colony-forming unit (CFU) assays and scored 14 days later. We detected an increased number of CFU-E (colony-forming unit erythroid) and CFU-GM (colony-forming unit granulocyte/macrophage) and a reduction of CFU-M (colony-forming unit macrophage) in iSAM-AGM samples cultured in the presence of DOX compared to the absence of DOX but no significant effect of DOX in SAM_NT samples (**Figure 3G** [↗](#)). These data indicate that activation of target transcription factors using our novel CRISPR strategy results in transcriptional remodelling and a steer in cell identity that we detected as a functional difference in the hematopoietic progenitor profile.

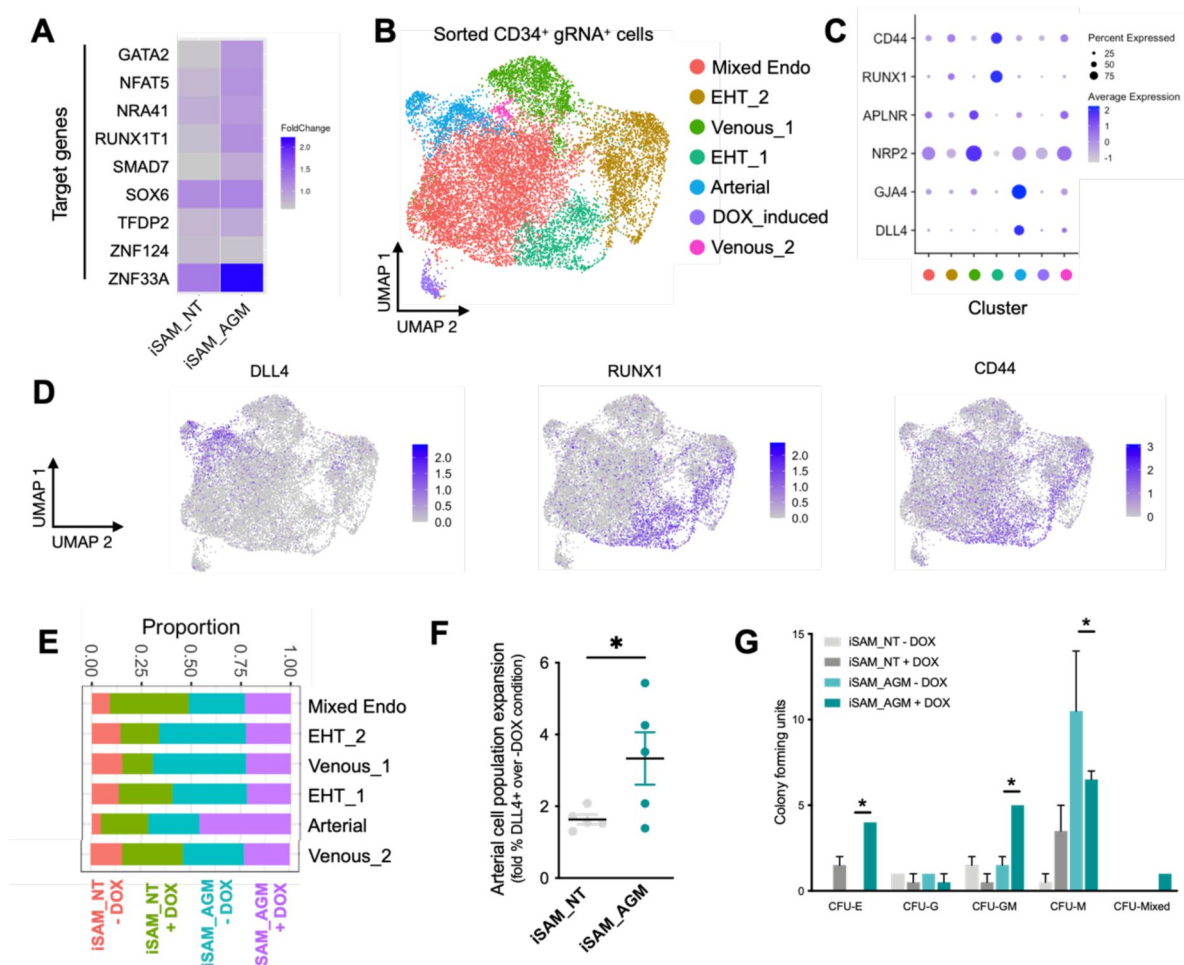


Figure 3

Single Cell RNA sequencing in combination with CRISPR activation identifies arterial cell type and functional hematopoietic expansion in association with activation of the 9 target genes.

A - Gene expression profile of target genes following target genes' activation, heatmap shows the expression level of the target genes in the iSAM_NT and iSAM_AGM treated with DOX following normalisation on the -DOX control. **B** - Dimension reduction and clustering analysis of the scRNAseq data following activation, filtered on cells where the gRNA expression was detected. **C** - Arterial (*GJA4*, *DLL4*), venous (*NRP2*, *APLNR*) and hemogenic marker (*CD44*, *RUNX1*) expression distribution in the clusters indicated by the colour. **D** - Expression distribution visualised on the UMAP plot showing the location of arterial cells marked by *DLL4*, and hemogenic endothelium marked by *CD44* and *RUNX1*. **E** - Contribution of the different libraries to the clusters showing that arterial cell cluster is overrepresented in the iSAM_AGM treated with DOX, compared to the other libraries. **F** - Expansion of the arterial population assessed by the membrane marker expression of DLL4⁺ following targets' activation, quantified by flow cytometry at day 8 of differentiation (Data are normalised on the iSAM_NT + DOX sample, n=5 independent differentiations, * p = 0.0417 paired t-test). **G** - Colony forming potential of the suspension progenitor cells derived from the two lines treated with or without DOX following OP9 coculture activation, data show the colony obtained for 104 CD34⁺ input equivalent (n=3 from independent differentiations * p<0.05, Tukey's two-way ANOVA).

The addition of IGFBP2 to the *in vitro* differentiation leads to a higher number of functional hematopoietic progenitor cells

To better understand the molecular mechanism behind the increased progenitor development, we compared the expression profile of the arterial cells between the different activation libraries, and we obtained a list of genes upregulated upon activation of the targets (Table 1). The most upregulated gene, *IGFBP2*, was expressed at significantly higher levels in the iSAM_AGM library in the presence of DOX compared to the others (Figure 4A, Table 1). We then compared the gRNA distribution in these arterial cells from the iSAM_AGM treated with DOX to that of arterial cells without activation. We observed a significant enrichment of the *RUNX1T1*-specific gRNAs (Table 1), indicating that the increased *IGFBP2* expression could be downstream of *RUNX1T1* activation. IGF Binding Protein 2 is a member of the family of IGF binding proteins and is thought to be secreted from cells where it then binds IGF1, IGF2, and other extracellular matrix proteins, modulating their function. To test if the increased frequency of functional hematopoietic progenitors was due to IGFBP2 paracrine signalling, presumably derived from the arterial cells, we supplemented the media with IGFBP2 at 100ng/ml after the induction of endothelial cell differentiation. To explore the role of IGFBP2 we employed the parental iPSCs line, SFCi55 from which the iSAM line was derived. We isolated CD34⁺ cells at day 8 and co-cultured them on OP9 cells in presence of IGFBP2 for one week, then tested for their hematopoietic clonogenic potential using CFU assays (Figure 4B). Cells treated with IGFBP2 showed a significant increase in the total number of haematopoietic CFU colonies compared to cells in control cultures. To assess whether IGFBP2 also had a paracrine effect on the production of arterial cells themselves, we analyzed the proportion of DLL4⁺ cells, but no difference was detected in the presence of IGFBP2 (Figure 4C). This observation supports our hypothesis that the change in hematopoietic progenitor production following activation in the iSAM_AGM line is associated with differential gene expression within the arterial cells rather than expansion of that cell population. We then focused on the characterization of the cells derived from the CD34⁺ cells after coculture with the OP9 in the presence of IGFBP2. Our results showed a significant expansion of the CD34⁺ and CD43⁺ cell populations further supporting our hypothesis (Figure 4D). To address the potential role of *IGFBP2* *in vivo*, we analysed single cell sequencing data from the human AGM at Carnegie stages 14 and 15.

IGFBP2 is highly expressed within the AGM niche by stromal and epithelial cells and, most importantly, highest in endothelial cells (Figure 4E). Together these data show that IGFBP2 addition results in increased hematopoietic blood production *in vitro* and that the endothelial compartment is the most likely source of IGFBP2 both *in vitro* and *in vivo*.

IGFBP2 remodels the metabolic profile of endothelial cells

Following the observation that IGFBP2 supports haematopoietic progenitor differentiation *in vitro* from human iPSCs we performed a time course single-cell RNA experiment of SFCi55 iPSCs differentiated in its presence. We isolated CD34⁺ cells and plated them in EHT culture on laminin²⁸, rather than on OP9 support, to assess exclusively the specific effects of IGFBP2- on differentiating iPSCs. FAC-sorted single/live cells from day 10 and 13 in the presence and absence of IGFBP2 were subjected to single-cell RNAseq. Our time course transcriptomic analyses showed that IGFBP2 induced a change in the transcriptome, specifically on day 13 (Figure 5A). A cluster of endothelial cells enriched for genes associated with the KEGG pathway of growth factor binding, was detected almost exclusively at day 13; these cells displayed a different transcriptional signature (as indicated by a shift in their position in the UMAP embedding) in the presence of IGFBP2.

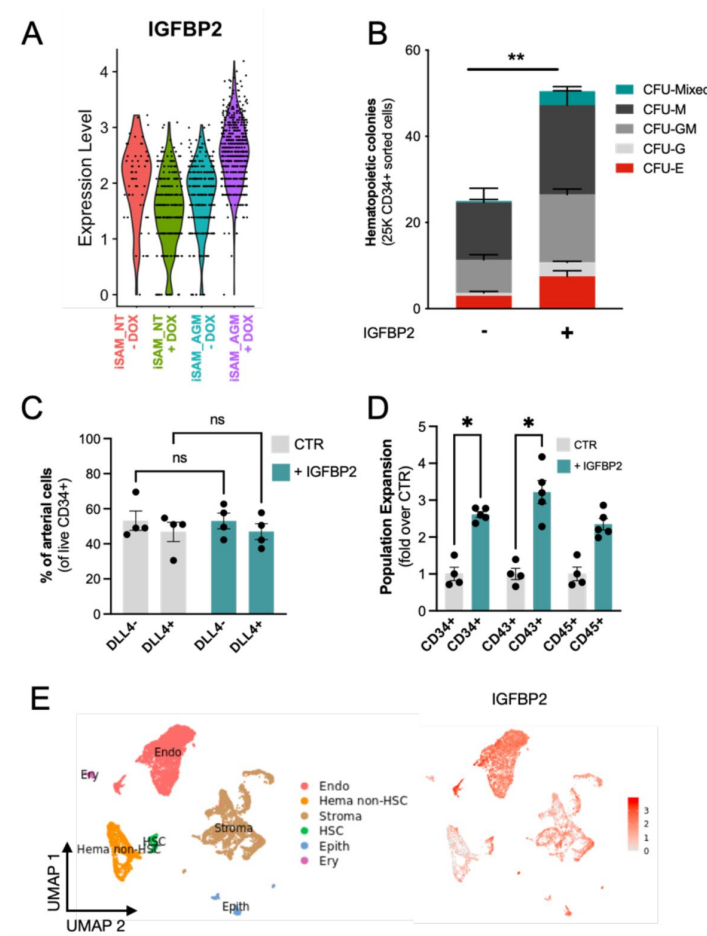


Figure 4

IGFBP2 addition to the *in vitro* differentiation leads to a higher number of functional hematopoietic progenitor cells.

A - Violin plot of *IGFBP2* expression profile in the arterial cells obtained from the different conditions, in the presence or absence of gRNAs and DOX. **B** - Number of hematopoietic colonies obtained after coculture on OP9 in presence or absence of *IGFBP2* (n=3-4 from independent differentiations, ** p=0.0080, Sidak's Two way ANOVA). **C** - Percentage of DLL4⁺ arterial cells differentiation within the CD34⁺ compartment analysed by flow cytometry in day 8 EBs (n=4 from independent differentiations, Two-way Anova, ns =p>0.99). **D** - Expansion of hematopoietic progenitors analysed using markers' expression on suspension progenitors derived after coculture of CD34⁺ cells onto OP9 support (data are expressed as fold over the CTR in the absence of *IGFBP2* (n=4 from independent differentiations, * p<0.02, Sidak's Two way ANOVA). **E** - Single-cell transcriptomic analysis of developing AGM collected from human embryos at Carnegie Stages 14 and 15 enriched for CD31⁺ and CD34⁺ showing the *IGFBP2* expression profile *in vivo* in the AGM.

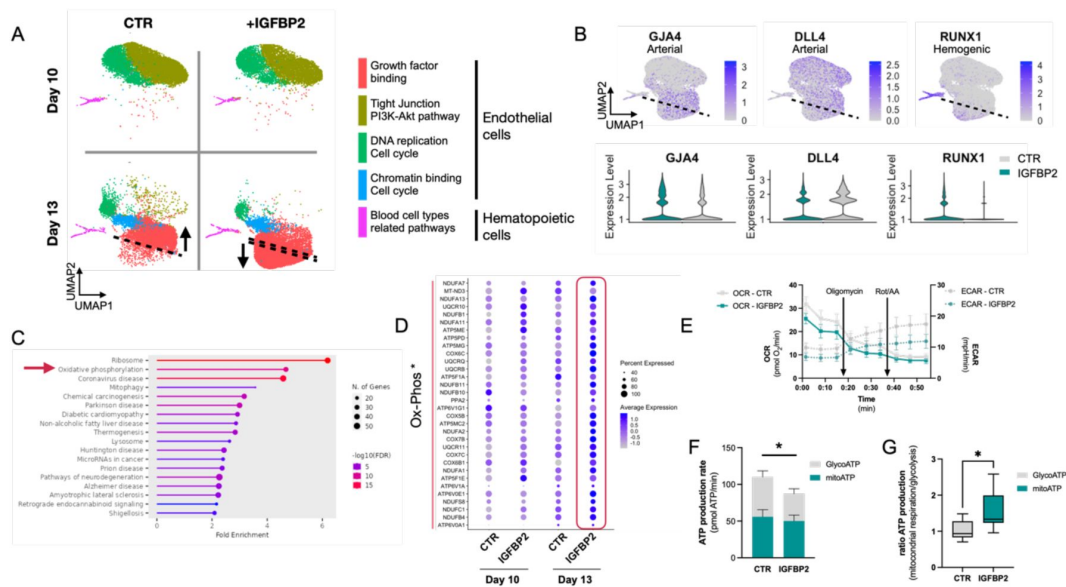


Figure 5

IGFBP2 alters cell metabolism by inducing a reduction in glycolytic ATP production.

A – Clustering analysis of the single cell transcriptomic time course analysis of differentiating cells at day 10 and day 13 in the absence (CTR) or presence of IGFBP2. Arrows indicate the difference in the clustering due to the addition of IGFBP2 compared to control. **B** – Expression profile of arterial markers, *GJA4* and *DLL4*, and hemogenic marker *RUNX1* (top - the dashed line shows the location of the shift in gene expression of cells treated with IGFBP2) and their expression profile in the endothelial cells cluster marked by Growth factor binding in absence (CTR) and in presence of IGFBP2. **C** – KEGG enrichment analysis of the genes upregulated at day 13 upon IGFBP2 treatment. The arrow shows the ranking of the Oxidative Phosphorylation pathway. **D** – Dot Plot showing the expression profile of the genes coding for the enzyme of the Oxidative Phosphorylation pathway. **E** – Oxygen Consumption Rate (OCR) and Extracellular Acidification Rate (ECAR) profile in cells at day 13 of differentiation reporting mitochondrial respiration and glycolysis, respectively. **F** – ATP production rate divided by that deriving from glycolysis and from mitochondrial respiration, in cells treated with IGFBP2 and controls at day 13. **G** – Ratio of the ATP production between glycolysis and mitochondrial respiration in cells treated with IGFBP2 and controls at day 13.

Interestingly, this cluster showed expression of *GJA4* across the cells, indicating their broad arterial identity, with a specific reduction of *DLL4* expression level concomitantly to an upregulation of *RUNX1* in the region of the cluster induced by IGFBP2 (**Figure 5B**). This increase in hemogenic identity within the endothelial cell compartment is consistent with the observation of increased functional progenitor production. We compared the transcriptome of cells at day 13 in the presence and absence of IGFBP2 and, using KEGG enrichment analysis, we observed that the genes that were upregulated by IGFBP2 were highly enriched in the oxidative phosphorylation term (**Figure 5C-D**). Since the metabolic switch between glycolytic to mitochondrial metabolism has been previously reported in definitive hematopoiesis^{29,30}, we tested whether the addition of IGFBP2 could result in a different ATP production profile. We analysed the ATP production at day 13, when *RUNX1* expression was induced by IGFBP2 addition, by using specific inhibitors of the complex I and II and complex V to quantify the intracellular ATP and mitochondrial synthesis, respectively. We detected a reduction in glycolytic-derived ATP (**Figure 5E-F**), which translates to a higher contribution of mitochondria metabolism for IGFBP2-treated cells (**Figure 5G**). We then looked at the expression levels of genes encoding the enzymes of the glycolytic pathway and its checkpoints, glucose and lactate transporters, monocarboxylate transporters and the enzymes associated with hexokinase and phosphofructose reactions. We observed a general downregulation of the glycolytic enzymes and its checkpoints, with the exception of *PFKM* and *SLC16A1*, (Figure S4), which were expressed at lower levels in the IGFBP2- treated cells at day 13 compared to the control on the same day. These differences were not observed on day 10 cells, which is consistent with our previous observations that the IGFBP2-induced remodelling of the transcriptome happens exclusively on day 13. These results indicate that the presence of IGFBP2 induces the upregulation of *RUNX1* and OxPhos genes and the metabolic remodelling of endothelial cells.

Discussion

The complexity and dynamism of developmental hematopoiesis *in vivo* have imposed challenges in accurately reproducing the process *in vitro*. We hypothesized that this could, in part, be due to the inability to recapitulate the appropriate transcriptional programme in cultured cells. Here, we developed a novel CRISPR-activation system to induce the expression of genes that are expressed at low levels in cells that are generated *in vitro*, to explore the downstream consequences of their activation and to assess the effects on emerging hematopoietic progenitor cells.

When we compared endothelial cells derived *in vitro* from hiPSCs to those in the AGM region of the human embryo, at the time point when early hematopoietic commitment takes place, we identified nine transcription factors that were expressed at a lower level in the cultured cells. Some of these had been associated with blood cell development, including *GATA2*^{31–33}, *SMAD7*³⁴, *NR4A1*³⁴, *SOX6*³⁵ and *RUNX1T1*^{15,17}. Other genes such as, *ZNF124*, *ZNF33A*, *NFAT5* and *TFDP2* had not been previously associated with hematopoiesis and could provide early evidence of their possible role in hematopoiesis that will require further studies. We were particularly interested in *RUNX1T1* (also known as ETO) which is associated with the leukemic fusion protein, AML1/ETO resulting from the t(8;21) chromosomal translocation³⁶. Furthermore, *RUNX1T1* expression has been recently detected in transcriptomic analyses of the human AGM region^{15,17}, but its precise role during the ontogeny of the blood system has not been elucidated. The addition of a capture sequence to the gRNA backbone enabled their detection coincidentally with the single-cell transcriptome, and this allowed us to demonstrate that *RUNX1T1* gRNAs were significantly enriched in cells within an expanded arterial cluster and their presence was associated with a high level of expression of *IGFBP2*. *IGFBP2* KO mice show increased expression of cell- cycle inhibitors and HSC apoptosis, implicating *IGFBP2* as a modulator of HSCs cell cycle and survival³⁷. More recently *IGFBP2* was reported as being highly expressed in the human AGM region at CS14 when HSCs are emerging¹⁷, supporting a possible role during developmental hematopoiesis but the precise molecular process was unclear. In this

study, we show that the addition of IGFBP2 recombinant protein in our *in vitro* model of EHT results in the emergence of an increased number of functional hematopoietic progenitors and we provide evidence to support the hypothesis that RUNX1T1 could regulated its expression. Because RUNX1T1 lacks a DNA binding domain, its direct involvement in the regulation of IGFBP2 expression must require association with other cofactors that are yet to be identified.

Our data, together with *in vivo* data¹⁷, show that IGFBP2 is expressed predominantly by endothelial cells. The supportive role of the endothelial niche in the development of HSCs has been studied *in vivo*^{38–40} and exploited *in vitro* to support HSCs emergence^{39,41}. We describe here a novel role for IGFBP2 in the remodelling of the metabolism of iPSC- derived endothelial cells and that this is associated with the induction of RUNX1, a hallmark of hemogenic fate. The addition of IGFBP2 in the culture induces upregulation of the genes of the oxidative phosphorylation pathway in association with an increased relative mitochondrial contribution to cellular ATP production. This is due to reduced glycolysis in association with the downregulation of genes coding for glycolytic enzymes and their checkpoints. The switch between glycolytic to mitochondrial metabolism has been shown to be essential for the EHT in definitive hematopoiesis^{29,30}. This switch is induced *in vivo* by mechanical cues downstream of the establishment of circulation, and it is required for functional HSCs development²⁹, while *in vitro*, the switch is driven by pyruvate mitochondrial catabolism, leading to definitive EHT as opposed to primitive³⁰. Further *in vivo* studies are required to characterize the role of IGFBP2 during AGM hematopoiesis to overcome the limitation of using an *in vitro* model such as iPSCs differentiation.

In conclusion, we detected transcriptional differences between *in vivo* and *in vitro* developmental hematopoiesis and developed a novel inducible gene activation system to identify novel molecular players during the endothelial-to-hematopoietic transition. Our multidisciplinary approach identified *RUNX1T1* and IGFBP2 as novel signalling molecules that support human blood progenitor development *in vitro*, the latter inducing a metabolic switch from cytoplasmic glycolysis to mitochondrial respiration. This study demonstrates that the combination of CRISPR-mediated activation of target genes with single-cell transcriptomic analysis in differentiating hPSCs can be a powerful approach to alter cell fate, providing a tool for gene function studies during human development. The fine epigenetic manipulation of the transcription can be readily applied to any cell lineage simply by adding specific gRNAs and it will be instrumental in exploring other developmental processes that can be, at least partially, be mimicked *in vitro* with human iPSCs.

Using this approach, we have identified the supportive role of paracrine IGFBP2, predominantly produced by endothelial cells, which induces transcriptional and metabolic remodelling in association with the induction of *RUNX1* expression.

Methods

Resource availability

R code is available at <https://github.com/afidanza/CRISPRa>. Raw and processed data have been deposited to ArrayExpress (E-MTAB-12748 – currently on hold). The AAVS1-iSAM are available on Addgene (Addgene #211495) and the gRNA 2.1 plasmids (Addgene #211496) (both currently on hold). Further information and requests for resources and reagents should be directed to and will be fulfilled by the corresponding author.

Pluripotent Stem Cells maintenance

hPSCs were maintained *in vitro* in StemPro hESC SFM (Gibco) with bFGF (R&D) at 20 ng/ml. Wells were coated with Vitronectin (ThermoFisher Scientific) at least 1 hour before plating and cells were passaged using the StemPro EZPassage tool (ThermoFisher Scientific). Media change was performed every day and cells passaged every 3–4 days at a ratio of 1:4.

Transfection

iPSCs SFCi55 and hESCs RUNX1-GFP were plated at 3×10^5 cells per a well of a 6 well plate and reverse transfected with 2 µg of DNA using the Xfect Transfection reagent (Clontech) and analyzed 2 days later.

HeLa cells were cultured in Dulbecco's Modified Eagle Medium/Nutrient Mixture F-12 (DMEM/F12) with Glutamax and 5% FCS (Gibco) and passaged every few days, at a ratio of 1:6. HEL were cultured in Iscove's Modified Dulbecco's Medium (IMDM) with 10% FCS (Gibco) and passaged every few days, at a ratio of 1:4. 2×10^5 cells were plated, transfected at 6–8 hours with 0.75 µg of DNA using Xfect Transfection reagent (Clontech) and then analysed 2 days after.

Immunocytochemistry

Cells were fixed in 4% PFA in PBS at room temperature for 10', permeabilized in PBS-T (0.4% Triton-X100) for 20' and blocked in PBS-T with 1% BSA and 3% goat serum for 1 hour. Primary antibodies were incubated in blocking solution over night at 4 °C (RUNX1 1:200 - ab92336, Abcam). Cells were then washed in PBS-T and incubated with secondary antibodies for 1 hour at room temperature (donkey α-rabbit 1:200 - A-11008 - Thermo Scientific). Cells were washed in PBS-T and counterstained with DAPI. Images were generated using the Zeiss Observer microscope.

Gene expression analysis

Total RNA was purified using the RNeasy Mini Kit (Qiagen) and cDNA synthesized from 500 ng of total RNA using the High Capacity cDNA synthesis Kit (Applied Biosystem). 2 ng of cDNA were amplified per reaction and each reaction was performed in triplicate using the LightCycler 384 (Roche) with SYBR Green Master Mix II (Roche). A melting curve was performed and analyzed for each gene to ensure the specificity of the amplification. *β-Actin* was used as reference genes to normalize the data ¹⁹.

Pluripotent Stem Cells differentiation to hematopoietic progenitors

hPSCs were differentiated in a xeno-free composition of SFD medium ¹⁴, BSA was substituted with human serum albumin, HSA (Irvine-Scientific). Day 0 differentiation medium, containing 10 ng/ml BMP4 was added to the colonies prior cutting. Cut colonies were transferred to a Cell Repellent 6 wells Plates (Greniner) to form embryoid bodies and cultured for two days. At day 2 media was changed and supplemented with 3 µM CHIR (StemMacs). At day 3, EBs were transferred into fresh media supplemented with 5 ng/ml bFGF and 15 ng/ml VEGF. At day 6 media was changed for final haematopoietic induction in SFD medium supplemented with 5 ng/ml bFGF, 15 ng/ml VEGF, 30 ng/ml IL3, 10 ng/ml IL6, 5 ng/ml IL11, 50 ng/ml SCF, 2 U/ml EPO, 30 ng/ml TPO, 10 ng/ml FLT3L and 25 ng/ml IGF1. From day 6 onward, cytokines were replaced every two days.

CD34 isolation

CD34+ cells were isolated using CD34 Magnetic Microbeads from Miltenyi Biotec, according to their manufacturing protocol. Briefly, Embryoid bodies were dissociated using Accutase (Life Technologies) at 37°C for 30'. Cells were centrifuged and resuspended in 150 µl of PBS + 0.5% BSA

+ 2mM EDTA with 50 μ l Fcr blocker and 50 μ l of magnetic anti-CD34 at 4°C for 30'. Cells were washed using the same buffer and transferred to pre-equilibrated columns, washed three times and eluted. After centrifugation, cells were resuspended in SFD media, counted and plated for OP9 coculture.

OP9 coculture and colony assay

OP9 cells were maintained in α -MEM supplemented with 20% serum (Gibco) and sodium bicarbonate (Gibco) and passaged with Trypsin every 3-4 days. The day before the co-culture, 45,000 OP9 cells were plated for each 12 well plates' well in SFD media. The day of the co-culture the 20,000 iSAM cells or 25,000 SFCi55 or H9 were plated in each well and culture in SFD media supplemented with 5 ng/ml bFGF, 15 ng/ml VEGF, 30 ng/ml IL3, 10 ng/ml IL6, 5 ng/ml IL11, 50 ng/ml SCF, 2 U/ml EPO, 30 ng/ml TPO, 10 ng/ml FLT3L and 25 ng/ml IGF1 and 100ng/ml IGFBP2. Cytokines were replaced twice during one week of coculture. At the end of the coculture, cells were collected by Trypsin and half of the well equivalent was plated in 2 ml of methylcellulose medium (Human enriched H4435, Stemcell Technologies). Cells were incubated in the assay for 14 days and then scored.

Laminin EHT culture

Laminin EHT culture was performed as previously described²⁸. Briefly, 24 well plates were coated for at least 2 hours with recombinant human Laminin-521 (Thermo-Fisher). Following CD34+ isolation at day 8, 400,000 CD34+ cells were seeded in each well of a pre-coated 24-well plate in SFD media supplemented with 5 ng/ml bFGF, 15 ng/ml VEGF, 30 ng/ml IL3, 10 ng/ml IL6, 5 ng/ml IL11, 50 ng/ml SCF, 2 U/ml EPO, 30 ng/ml TPO, 10 ng/ml FLT3L and 25 ng/ml IGF1 and 100ng/ml IGFBP2.

Flow cytometry staining and cell sorting

Embryoid bodies were dissociated using Accutase (Life Technologies) at 37°C for 30'. Cells were centrifuged and resuspended in PBS + 0.5% BSA + 2mM EDTA, counted and stained at 10^5 cells for a single tube. Cells were stained with antibodies for 30' at room temperature gently shaking. Flow cytometry data were collected using DIVA software (BD). For the sorting experiments, cells were stained at 10^7 cells/ml in presence of the specific antibodies. Sorting was performed using FACSARIA Fusion (BD) and cells were collected in PBS + 1% BSA. Data were analysed using FlowJo version 10.4.2.

Flow cytometry antibodies

For flow cytometry 10^5 cells per test were stained in 50 μ l of staining solution with the following antibodies: CD34 Percp-Efluor710 (4H11 eBioscience, 1:100), CD34 Pe (4H11 eBioscience, 1:200), CD43 APC (eBio84-3C1, 1:100), CD45 FITC (2D1 ebioscience, 1:100), DLL4 Pe (MHD4-46 Biolegend, 1:200)

iSAM plasmid generation

The iSAM plasmid was obtained by Gibson assembly of four fragments. The first fragment, the backbone, was a DOX-inducible AAVS1 targeted plasmid expressing an E6-E7-IRES- ZsGreen which was excised by BstBI and NdeI. The second fragment, one of the adapters, was derived from the UniSAM plasmid that we previously generated (Addgene #99866) by PCR with the following primer sets

FW_ aggggacccggttcgagaaggggctctcatcactagggccgtagctctagagagcgtcgaatt,
RV_ ttcgggtcccaattgccgtcgtctggcggtcttccaccttctcttcttggggctcatggtggcc. The UniSAM cassette was obtained also from the UniSAM plasmid via digestion with BsrGI and BsiWI. Finally, the last fragment consisting of another adapter for the Gibson was custom synthesised and contained

overlapping sequences flanking a chicken b-globin insulator that we inserted to prevent cross-activation of the EF1 α -promoter and the TRE-GV promoter driving the iSAM. Correct assembly was verified by Sanger sequencing.

iSAM cell lines derivation

The iSAM plasmid was used together with ZNFs specific for the AAVS1 locus to mediate specific integration in SFCi55 human iPSCs line ¹⁴[\[20\]](#). Briefly, 10 μ g of AAVS1-iSAM with 2.5 μ g of each ZNFs, left and right, using Xfect (Takara) according to the manufacturer protocol. Cells were selected using Neomycin. Single clones were picked, amplified, and initially screened by mCherry expression upon DOX addition. Clones that expressed the fluorescent tag were screened for specific integration using PCR followed by Sanger sequencing for the correctly integrated clones. 100 ng of genomic DNA was amplified using the EmeraldAmp[®] MAX HS Takara and specific primer sets (Table 1). For the specific AAVS1 integration site, Sigma_AAVS1 - CGG AAC TCT GCC CTC TAA CG and NeoR -GAT ATT GCT GAA GAG CTT GGC GG were used with the PCR conditions of 95 °C for 7 min, 32 cycles of 95 °C for 15 s, 57 °C for 30 s, and 72 °C for 1 min, with the final elongation step at 72 °C for 7 min. For the Wild type locus screening, Sigma_AAVS1 - CGG AAC TCT GCC CTC TAA CG and AAVS1_EXT3_RV – ACA CCC AGA CCT GAC CCA AA were used with the PCR cycling conditions of 95 °C for 7 min, 30 cycles of 95 °C for 15 s, 57 °C for 30 s, and 72 °C for 2 min, with the final elongation step at 72 °C for 6 min.

Capture sequencing addition to the gRNA backbone

The Capture sequence 2 was added to the gRNA_Puro_Backbone (Addgene #73797) by PCR. Briefly, the capture sequence was added before the termination signal of the gRNA followed by a BamHI site using the following PCR primers:

gRNA_FW gagggcctatttcccatgattct,

gRNA_Cap_RV aaaaaaggatccaaaaaCCTTAGCCGCTAATAGGTGAGCgcaccgactcggtgcc.

The gRNA backbone was replaced from the original plasmid via NdeI and BamHI digestion, followed by ligation of the PCR produced following the same digestion. Correct integration of the insert was verified by Sanger sequencing.

Target genes identification

Candidate genes were identified by comparing *in vivo* hemogenic endothelium ¹⁵[\[21\]](#) (GSE135202) and *in vitro* iPSCs-derived endothelial cells that we previously generated ¹⁴[\[20\]](#) (E-MTAB-9295). Briefly, the two datasets were merged and normalised using the R package Seurat. Specific markers for hemogenic endothelium were identified and transcription factors were sorted based on their GO annotation. Within those genes we filtered those detected in more than 50% of the *in vivo* hemogenic endothelium and expressed in less than 25% of the *in vitro* derived endothelial cells. This pipeline identified 9 target genes *RUNX1T1*, *NR4A1*, *GATA2*, *SMAD7*, *ZNF124*, *SOX6*, *ZNF33A*, *NFAT5*, *TFDP*.

AGM gRNAs library preparation

sgRNA design was performed by selecting the top candidates for on-target and off-target score. Between 5 and 7 guides per gene were designed for *RUNX1T1*, *NR4A1*, *GATA2*, *SMAD7*, *ZNF124*, *SOX6*, *ZNF33A*, *NFAT5*, *TFDP2* using the CRISPRpick tool from the Broad Institute (<https://portals.broadinstitute.org/gppx/crispick/public> ⁴²[\[22\]](#)) (Table S1). All the guide variants were Golden Gate cloned with the gRNA 2.1 backbone according to the established protocol ⁴²[\[22\]](#). The 49 plasmids were pooled together in an equimolar ratio and the library prep was subsequently used to produce lentiviral particles with a second-generation production system. Briefly, the psPAX2 packaging plasmid, pMD2.G envelope, and the AGM vector library were co-transfected using

polyethyleneimine (PEI) (Polysciences, Warrington, PA, USA) as previously detailed ⁴³, Lentiviral particles-containing supernatants were harvested 48–72 h post-transfection, concentrated by ultracentrifugation and titered in hiPSCs cells.

iSAM_AGM and iSAM_NT cell line derivation

The selected iSAM clone (3.13 internal coding) was infected with viral particles containing either the AGM library or the non targeting gRNA (NT) at a MOI of 10. The iSAM cells were plated the afternoon before at 7×10^6 cells into a T125 in the presence of 10 μ M Rock Inhibitor (Merk) which was maintained until the day following the infection. Cells were infected in presence of 8 μ g/ml of Polybrene (Merk). Puromycin selection was initiated 36 hours post-infection and maintained during their culture until the beginning of the differentiation. Both lines were tested for integration of the gRNAs. Briefly, 100 ng of isolated gDNA was amplified using the PrimeSTAR MAX PCR mix (Takara) using the primers gRNA_screening_FW and gRNA_screening_RV (Table1). Purified amplicons were subjected to Sanger sequencing.

Single Cell RNA sequencing

For the iSAM_AGM and iSAM_NT single cell RNA sequencing experiment, embryoid bodies obtained from day 10 of differentiation were dissociated using Accutase (Life Technologies') at 37°C for 30'. For the IGFBP2 experiment, day 10 and day 13 cells from the Laminin EHT culture were detached from the adherent layer using Accutase (Life Technologies') at 37°C for 5'. From both experiments, cells were centrifuged and resuspended in CD34-Pe staining solution at a density of 10^7 /ml. CD34+/live/single cells were FAC-sorted in PBS + 0.1 % BSA. Cell viability was also confirmed by Trypan blue stain for an accurate count. Around 15000 cells per sample were loaded into the 10X Chromium Controller, and single cell libraries were obtained using the Chromium single cell 3' Reagent Kits v3 (10XGenomics) according to manufacturer protocol. The four libraries were indexed using SI PCR primers with different i7 indexes to allow for demultiplexing of the sequencing data. RNA concentration was obtained using Qubit RNA HS (Thermo-Fisher). Quality of the obtained libraries were verified using LabChip GX (PerkinElmer). Libraries were sequenced using NextSeq 2000 technology (Illumina) at 50.000 reads/cell. Data were aligned to GRCh38 using the Cell Ranger dedicated pipeline (10XGenomics). Data filtering, dimension reduction, clustering analysis, differentially expressed genes and cell cycle analysis were obtained using Seurat R package (version 4.1.0)⁴⁴. Pseudotemporal ordering was performed using Monocle 3 R package ⁴⁵. KEGG pathways was performed using ShinyGo⁴⁶. The code is available on Github at <https://github.com/afidanza/CRISPRa>, the raw data have been submitted to Array Express (E-MTAB-12748), and the browsable processed data will be made available at the time of publication on our website containing previous sequencing data at <https://lab.antonellafidanza.com>.

ATP production analysis

At day 8 of differentiation, 20.000 CD34+ were plated for each precoated well of the Seahorse XFp mini Cell Culture Plates (Agilent) precoated with rhLaminin-521 (Thermo- Fisher). At day 13, the cells' ATP production rate was analysed with the Seahorse XF Real- Time ATP Rate Assay kit (Agilent) according to manufacturer protocol. Briefly, the media was changed prior to the assay for the XF DMEM medium, pH 7.4, supplemented with 10 mM Seahorse XF glucose, 1mM Seahorse XF pyruvate and 2mM Seahorse XF Glutamine (Agilent) and incubated for 45-60 minutes in a non-CO2 incubator. Oligomycin and Rotenone/AA solutions were prepared and added in the cartridge and finally loaded together with the cells in the Seahorse XF Mini Analyzer using the dedicated software. Data were collected at the end of the run, and the values of ATP production were calculated according to the ATP Production Rate Calculation provided by Agilent.

Author contribution

AF designed the study, performed experiments and bioinformatic analysis, wrote the paper and led the research. PP, TV, FPL, HM performed experiments. AM, HT, MS provided support to the experiments. NR performed bioinformatic analysis. LF and PM helped the design of the study and the research. All authors provided essential feedback on the experiments and the manuscript.

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Reviewer #1 (Public Review):

Summary:

The work from Petazzi et al. aimed at identifying novel factors supporting the differentiation of human hematopoietic progenitors from induced pluripotent stem cells (iPSCs). The authors developed an inducible CRISPR-mediated activation strategy (iCRISPRa) to test the impact of newly identified candidate factors on the generation of hematopoietic progenitors in vitro. They first compared previously published transcriptomic data of iPSC-derived hematopoietic populations with cells isolated ex vivo from the aorta-gonad-mesonephros (AGM) region of the human embryo and they identified 9 transcription factors expressed in the aortic hemogenic endothelium that were poorly expressed in the in vitro differentiated cells. They then tested the activation of these candidate factors in an iPSC-based culture system supporting the differentiation of hematopoietic progenitors in vitro. They found that the IGF binding protein 2 (IGFBP2) was the most upregulated gene in arterial endothelium after activation and they demonstrated that IGFBP2 promotes the generation of functional hematopoietic progenitors in vitro.

Strengths:

The authors developed an extremely useful doxycycline-inducible system to activate the expression of specific candidate genes in human iPSC. This approach allows us to simultaneously test the impact of 9 different transcription factors on in vitro differentiation of hematopoietic cells, and the system appears to be very versatile and applicable to a broad variety of studies.

The system was extensively validated for the expression of 1 transcription factor (RUNX1) in both HeLa cells and human iPSC, and a detailed characterization of this test experiment was provided.

The authors exhaustively demonstrated the role of IGFBP2 in promoting the generation of functional hematopoietic progenitors in vitro from iPSCs. Even though the use of IGFBP2-interacting proteins IGF1 and IGF2 have been previously reported in human iPSC-derived hematopoietic differentiation in vitro (Ditadi and Sturgeon, *Methods* 2016; Ng et al., *Nature Biotechnology* 2016), and IGFBP-2 itself has been shown to promote adult HSC expansion ex vivo (Zhang et al., *Blood* 2008), its role on supporting in vitro hematopoiesis was demonstrated here for the first time.

Weaknesses:

Although the authors performed a very thorough characterization of the system in proof-of-principle experiments activating a single transcription factor, the data provided when 9 independent factors were used is not sufficient to fully validate the experimental strategy.

Indeed, in the current version of the manuscript, it is not clear whether the results presented in both the scRNAseq analysis and the functional assays are the consequence of the simultaneous activation of all 9 TF or just a subset of them. This is essential to establish whether all the proposed factors play a role during embryonic hematopoiesis, and a more complete analysis of the scRNAseq dataset could help clarify this aspect.

Similarly, the data presented in the manuscript are not sufficient to clarify at what stage of the endothelial-to-hematopoietic transition (EHT) the TF activation has an impact. Indeed, even though the overall increase of functional hematopoietic progenitors is fully demonstrated, the assays proposed in the manuscript do not clarify whether this is due to a specific effect at the endothelial level or to an increased proliferation rate of the generated hematopoietic progenitors. Similar conclusions can be applied to the functional validation of IGFBP2 in vitro.

The overall conclusions are sometimes vague and not always supported by the data. For instance, the authors state that the CRISPR activation strategy resulted in transcriptional remodeling and a steer in cell identity, but they do not specify which cell types are involved and at what level of the EHT process this is happening. In the discussion, the authors also claim that they provided evidence to support that RUNX1T1 could regulate IGFBP2 expression. However, this is exclusively based on the enrichment of RUNX1T1 gRNA in cells expressing higher levels of IGFBP2 and it does not demonstrate any direct or indirect association of the two factors.

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Reviewer #2 (Public Review):

To enable robust production of hematopoietic progenitors in-vitro, Petazzi et al examined the role of transcription factors in the arterial hemogenic endothelium. They use IGFBP2 as a candidate gene to increase the directed differentiation of iPSCs into hematopoietic progenitors. They have established a novel induced-CRISPR mediated activation strategy to drive the expression of multiple endogenous transcription factors and show enhanced production of hematopoietic progenitors through expansion of the arterial endothelial cells. Further, upregulation of IGFBP2 in the arterial cells facilitates the metabolic switch from glycolysis to oxidative phosphorylation, inducing hematopoietic differentiation. While the overall study and resources generated are good, assertions in the manuscript are not entirely supported by the experimental data and some claims need further experimental validation.

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Author response:

We thank both reviewers for their feedback and for underlying the potential of our new tool and experimental approach to identify signalling molecules that can improve the in vitro derivation of specific cell types from human pluripotent stem cells. To address the reviewers' points we plan to carry out further analysis that should solidify our conclusions. We will also edit the text to temper conclusions where appropriate.